MIL-F-2150C (AR)
Amendment 4
8 October 1981
SUPERSEDING
Amendment 3
20 September 1971

MILITARY SPECIFICATION

FUZE, PD, M503A2
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-F-2150C (MU), dated 20 September 1968 and is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the following preamble underneath the title:

"This specification is approved by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense."

PAGE 2

* 3.5: Delete in its entirety and substitute the following:

"3.5 Air leakage. The Loaded Fuze Assembly (dwg. 9215031) shall be capable of withstanding an air pressure of 5 psi for one minute with a pressure drop no greater than 0.10 psi, max."

3.6.1, line 3: Delete "100" and substitute "400."

* 3.6.2: Delete in its entirety.

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* 3.6.3: Delete in its entirety.

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* 4.2.1: Delete in its entirety and substitute the following:

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"4.2.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 90 assemblies and 50 complete sets of parts (i.e. 50 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and fulfilling the contract. The ninety (90) complete end item fuzes are to be tested as follows:

a. 30 complete fuzes - Jolt & Jumble as per para 4.4.1.

b. 30 complete fuzes - Transportation Vibration as per para 4.4.2 and after T.V. Testing, marked and included in ballistic test sample "d".

c. 30 complete fuzes - Waterproofness Testing as per para 4.4.3.

d. 30 complete rounds - Ballistic Testing as per para 4.4.7.

All parts and materials, including packaging and packing, shall be obtained from the same source of supply as well as be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assume that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification, such as to significantly affect product uniformity as determined by the Government."

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* 4.3.1: Add "b." as follows:

"b. Booster and lead assemblies each of one lot interfix number from one manufacturer."

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4.3.2.1, Critical defect 1: Delete "Visual" and substitute "test (see para 4.3.3.8)."

4.3.2.2: Delete in its entirety.

4.3.2.3: Delete in its entirety.

4.3.2.4, Critical defect 1: Delete "Visual" and substitute "Gage."

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* 4.3.3.3: Delete in its entirety and substitute the following:

"4.3.3.3 Air Leakage (see 3.5). Operability deficiency Major defect - Code No. 07011. Fifty Loaded Fuze Assemblies (dwg. 9215031) shall be randomly selected from each lot and tested as specified in 4.4.12. If one or more of the samples fail to comply with the requirements, the lot shall be rejected."

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4.3.3.4: Delete in its entirety.

4.3.3.5: Delete in its entirety.

4.3.3.6: Delete in its entirety.

* 4.3.3.7:

Lines 2 and 3: Delete "non-functioning failure - Critical defect - Code No. 07008."

Line 5: Delete "through 4.3.3.7.3."

* 4.3.3.7.1: Delete in its entirety and substitute the following:

"4.3.3.7.1 Ballistic screen functioning test. A sample size of 30 shall be randomly selected from each lot and fired against a 1 inch screen at a distance of 400 ft. from

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the muzzle of the weapon. The fuze shall not premature function (see 6.5) and shall cause the projectile to function high order upon impact against a target with an acceptance criteria of accept on 2, reject on 3."

4.3.3.7.1.1: Delete in its entirety.

4.3.3.7.1.2: Delete in its entirety.

* 4.3.3.7.2: Delete in its entirety.

4.3.3.7.2.1: Delete in its entirety.

4.3.3.7.2.2: Delete in its entirety.

* 4.3.3.7.3: Delete in its entirety.

4.3.3.7.3.1: Delete in its entirety.

4.3.3.7.3.2: Delete in its entirety.

* Table I: Delete in its entirety.

4.3.3.8: Delete in its entirety and substitute the following:

"4.3.3.8 (see dwg. 9234644) Static Non-Arm Test. The fuze shall be tested as received less pellets, cup booster and cup booster lead. Critical defect, Code No. 01001. This test shall be performed one hundred (100) percent. The fuze shall not be spun when subjected to this test. Any one (1) assembly which fails to pass this test shall be classed defective. The rejected fuzes shall be subject to failure analysis. The accuracy of the inspection and inspection equipment shall be verified hourly using standards containing critical defects. These standards should be readily identifiable to ensure their recovery, but should be constructed in such a manner that they are not easily recognized by cognizant inspection personnel responsible for their detection. In case the standard is missed and not classed as a reject, all material inspected and accepted since the last acceptable standard verification check shall be subjected to reinspection. The test shall be performed

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as specified in 4.4.8 using equipment in accordance with 4.3.4. This equipment shall be capable of automatically segregating accepted product from rejects by sensing the presence or absence of required phosphorescent paint while not being dependent on operator observation."

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* 4.4.3: Delete in its entirety and substitute the following:

"4.4.3 Air leakage. The assembly shall be placed in a suitable test fixture. A steady state air pressure of 5 psi shall be applied for one minute. Any assembly which exhibits a pressure drop greater than 0.10 psi during the one minute shall be classed defective."

4.4.4: Delete in its entirety.

4.4.5: Delete in its entirety.

4.4.6: Delete in its entirety.

* 4.4.7.1: Delete in its entirety.

PAGE 13

* 4.4.7.2: Delete in its entirety and substitute the following:

"4.4.7.2 Ballistic screen functioning. The ballistic sample shall be assembled to a 57MM Ctg, M306A1 (with HE or spotting charge) and fired from the 57MM recoilless rifle loaded to standard service velocity and pressure. The sample shall be fired at a one inch thick plywood bursting screen located 400 feet from the muzzle of the weapon. No more than 8 rounds shall be fired on each 4 inch x 8 inch target. Rounds that strike the target leaving a complete hole with a minimum of 2 inches of solid target material from the edge of a previous hole or edge of target shall be counted. All rounds less than this minimum shall be discounted."

* 4.4.7.3: Delete in its entirety.

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Add new paragraph 4.4.8 as follows:

"4.4.8 Static non-arming test. Each fuze shall be tested as specified in 4.3.3.8 for the non-arm condition (safe) as evidenced by the presence of phosphorescent green paint in the scribe lines. The fuzes shall be passed through the test equipment which will sense and segregate and safe non-arm fuzes from those in the armed condition."

5.1: Delete "packing" and substitute "packaging."

5.1.1: Delete "Level A" and substitute "Level C".

5.2.1: Delete "Level A" and substitute "Level C".

"The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

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